

NCSA CTF Boot Camp #2

Digital Forensics

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Confidentiality class: Public



Disclaimer

- จุดประสงค์ของการบรรยาย นี้เพื่อแบ่งปันความรู้ ทางด้านความปลอดภัยระบบสารสนเทศ
- ไม่สนับสนุนการนำความรู้ทางด้านความปลอดภัยฯ ไปใช้ในทางที่ผิดกฎหมายทั้งหมด
- ตัวอย่างโค้ด และรูปใน การบรรยาย นี้ เป็นระบบจำลองของทางผู้บรรยาย ไม่ใช่ระบบลูกค้า



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Digital Forensics คืออะไร

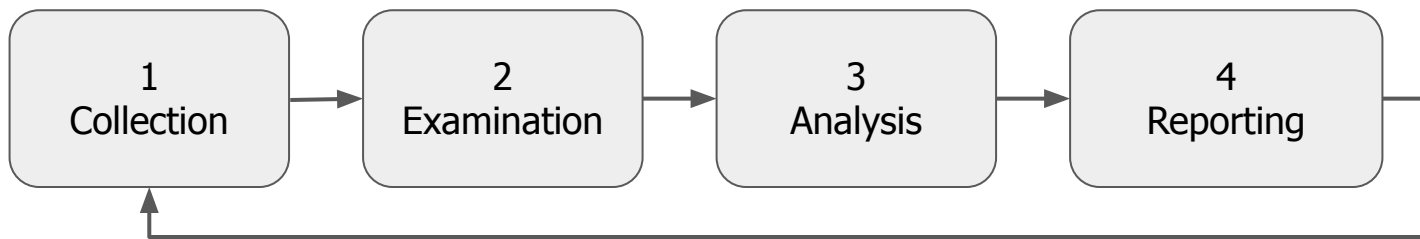


เป็นกระบวนการเก็บรวบรวม วิเคราะห์ และ
รักษาหลักฐานที่ได้จากระบบดิจิทัล เพื่อนำ
มาใช้ในการสืบสวนหรือดำเนินคดีทาง
กฎหมาย

หลักฐานเหล่านี้อาจมาจาก

- อุปกรณ์คอมพิวเตอร์
- โทรศัพท์มือถือ
- เครือข่าย
- หรืออุปกรณ์จัดเก็บข้อมูลอื่น ๆ

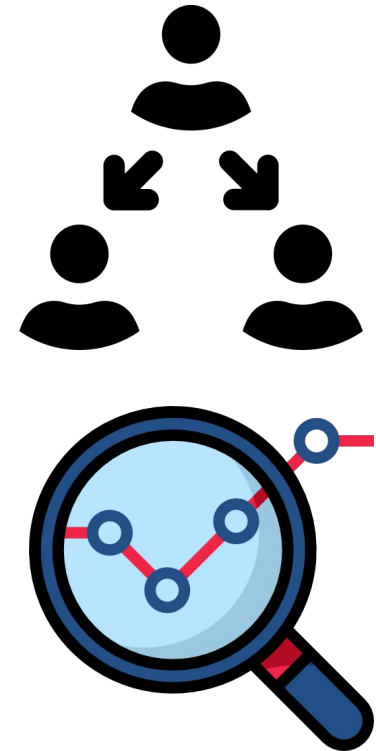
The Process of Digital Forensics



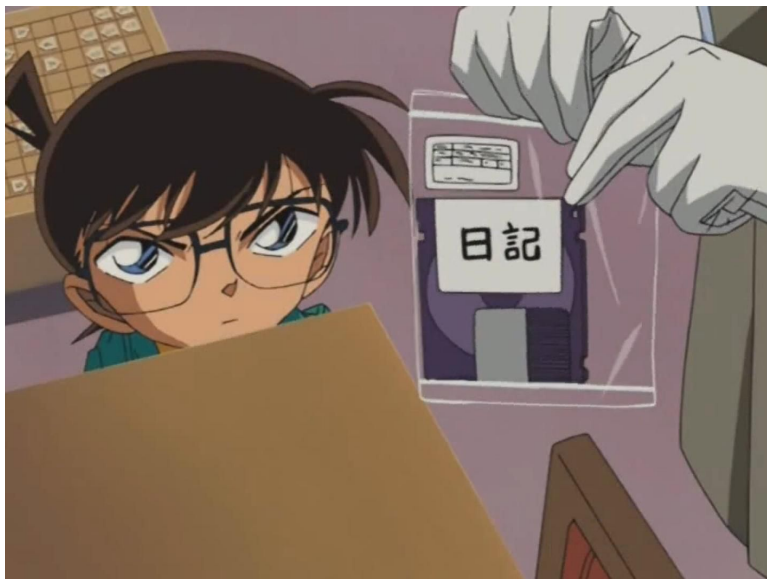
Media → Data → Information → Evidence

NIST SP 800-86 - Digital Forensics
Flow Standard

Digital Forensics



Digital Forensics (Real World) VS (CTF)



Detective Conan" The Dark Footprint: Part 2"

ประเภทของการ Digital Forensics

- Log Analysis
- File Analysis
- Memory Forensics
- Disk Forensics
- Malware Analysis
- etc.

ดาวน์โหลดไฟล์แลป

https://bootcamp.p7z.pw/4_for/

← → ↻ 🔒 🔓 https://bootcamp.p7z.pw/4_for/ ☆ 📧 ⬇️ 👤 📄		
Index of /4_for/		
../		
for_lab1_access.log	14-Sep-2024 02:44	29972
for_lab2_cat.jpg	14-Sep-2024 02:44	878121
for_lab3_verify.zip	14-Sep-2024 02:44	81899



ทำความเข้าใจ Log

Log File

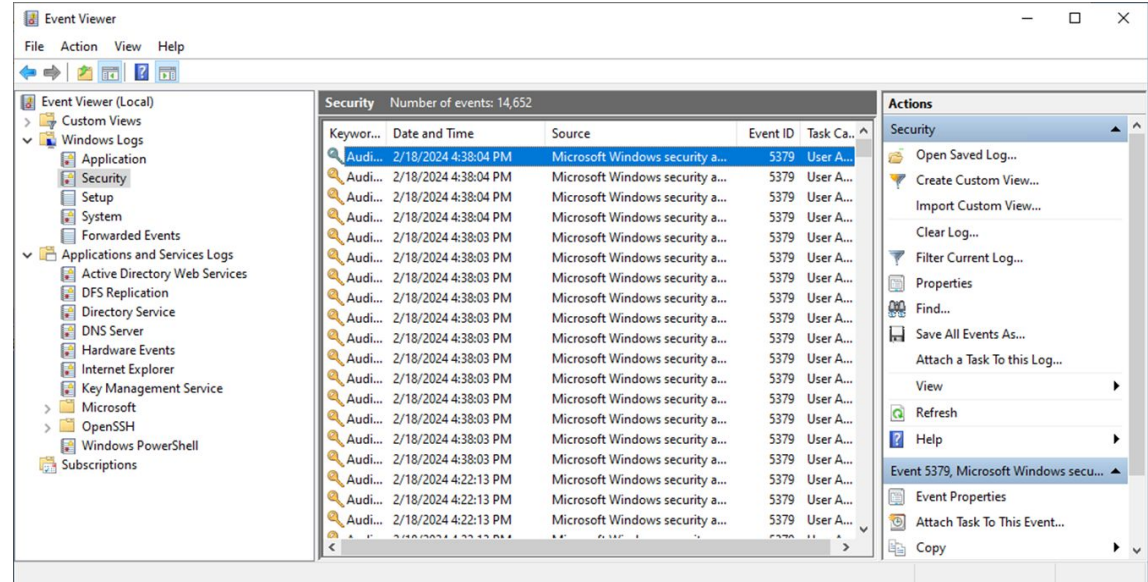
- Application Log
- Security Log
- Server Log
- System Log
- Network Log
- Database Log

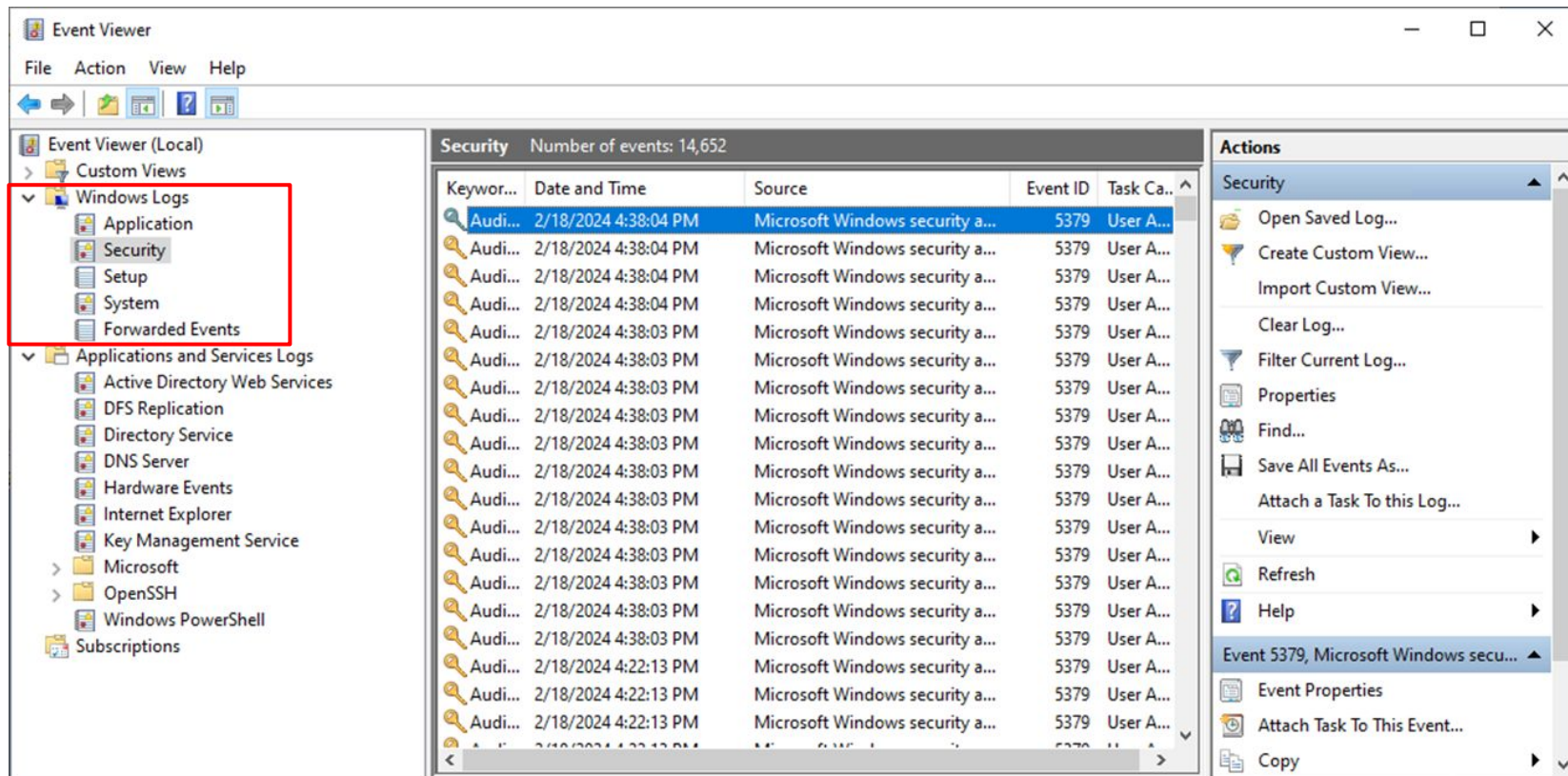
```
192.168.1.1 - - [01/Sep/2024:12:00:00 +0000] "GET /index.html HTTP/1.1" 200 1024 "https://example.com/home" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/100.0.4896.127 Safari/537.36"
192.168.1.2 - - [01/Sep/2024:12:05:30 +0000] "POST /login HTTP/1.1" 302 512 "https://example.com/login" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Firefox/89.0"
192.168.1.3 - - [01/Sep/2024:12:10:15 +0000] "GET /about.html HTTP/1.1" 200 2048 "https://example.com/home" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:90.0) Gecko/20100101 Firefox/90.0"
192.168.1.4 - - [01/Sep/2024:12:15:45 +0000] "GET /images/logo.png HTTP/1.1" 200 4096 "https://example.com/index.html" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Safari/537.36"
192.168.1.5 - - [01/Sep/2024:12:20:30 +0000] "GET /contact.html HTTP/1.1" 404 256 "https://example.com/home" "Mozilla/5.0 (Linux; Android 10; Pixel 4 XL) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.77 Mobile Safari/537.36"
```


The Path of Log File on Windows

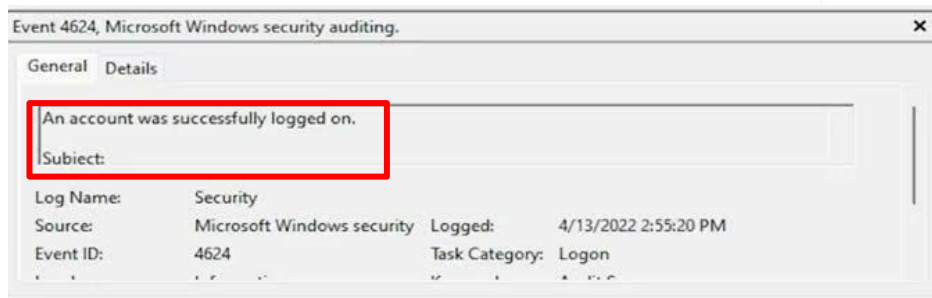
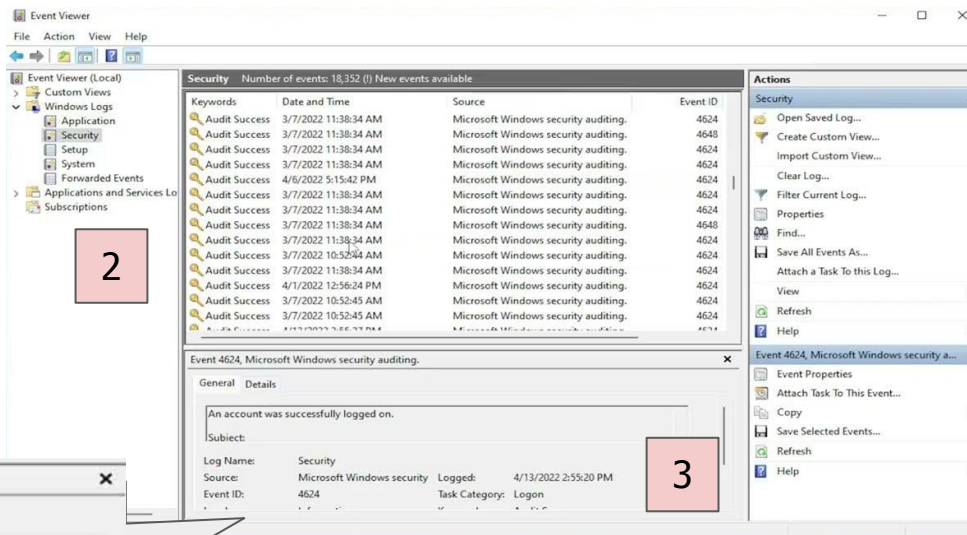
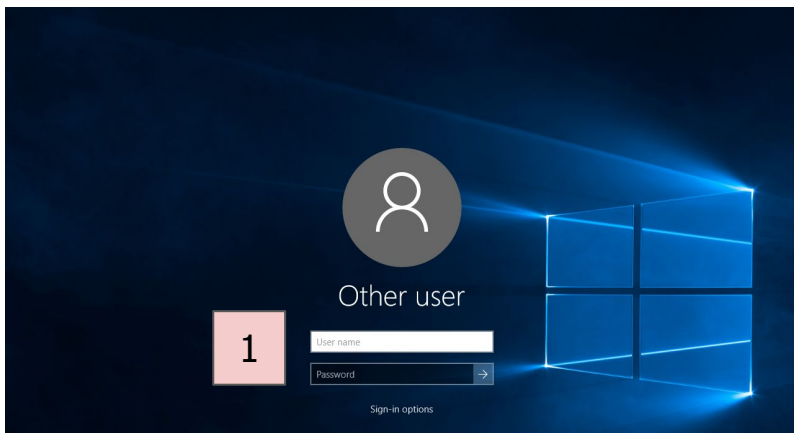
บน Windows ไฟล์ Log มักถูกจัดการผ่าน Event Viewer และจะถูกเก็บในรูปแบบของ Event Logs และเก็บไว้ที่ %SystemRoot%\System32\Winevt\Logs\ เช่น C:\Windows\System32\Winevt\Logs\

1. Application Log
2. System Log
3. Security Log
4. Setup Log
5. Forwarded Events Log





Windows Security Log



The Path of Log Files on Linux

บน Linux ไฟล์ Log มักถูกเก็บไว้ที่ **/var/log/**

1. syslog
2. auth.log
3. boot.log
4. cron.log
5. apache2/ หรือ httpd/
6. mysql/ หรือ mariadb/
7. kern.log

```
(kali㉿kali)-[/var/log]
└─$ ls -la
total 1596
drwxr-xr-x 21 root      root      4096 Sep  7 12:34 .
drwxr-xr-x 12 root      root      4096 Sep  4 19:34 ..
lrwxrwxrwx  1 root      root        39 Sep  4 12:24 README → ../../usr/s
hare/doc/systemd/README.logs
-rw-r--r--  1 root      root      22869 Sep  7 12:35 Xorg.0.log
-rw-r--r--  1 root      root     21855 Sep  7 12:35 Xorg.1.log
-rw-r--r--  1 root      root    82365 Sep  4 12:29 alternatives.log
drwxr-x--  2 root      adm       4096 Sep  4 12:28 apache2
drwxr-xr-x  2 root      root       4096 Sep  4 12:55 apt
-rw-----  1 root      root       6531 Sep  4 19:34 boot.log
-rw-rw----  1 root      utmp         0 Sep  4 12:24 btmp
-rw-r--r--  1 root      root    1327536 Sep  4 12:55 dpkg.log
-rw-r--r--  1 root      root         0 Sep  4 12:24 faillog
-rw-r--r--  1 root      root       7354 Sep  4 12:31 fontconfig.log
drwxr-xr-x  2 _gvm      _gvm       4096 Apr 23 10:32 gvm
drwx-----  3 inetsim   inetsim     4096 Sep  4 12:29 inetsim
drwxr-xr-x  3 root      root       4096 Sep  4 12:34 installer
drwxr-sr-x+  3 root      systemd-journal 4096 Sep  4 19:34 journal
```

/var/log

```
(kali㉿kali)-[/var/log]
└─$ ls -la
total 1596
drwxr-xr-x 21 root root      4096 Sep  7 12:34 .
drwxr-xr-x 12 root root      4096 Sep  4 19:34 ..
lrwxrwxrwx  1 root root        39 Sep  4 12:24 README → ../usr/s
hare/doc/systemd/README.logs
-rw-r--r--  1 root root     22869 Sep  7 12:35 Xorg.0.log
-rw-r--r--  1 root root    21855 Sep  7 12:35 Xorg.1.log
-rw-r--r--  1 root root    82365 Sep  4 12:29 alternatives.log
drwxr-xr-x  2 root adm       4096 Sep  4 12:28 apache2
drwxr-xr-x  2 root root      4096 Sep  4 12:55 apt
-rw-r----- 1 root root     6531 Sep  4 19:34 boot.log
-rw-rw----- 1 root utmp          0 Sep  4 12:24 btmp
-rw-r--r--  1 root root   1327536 Sep  4 12:55 dpkg.log
-rw-r--r--  1 root root          0 Sep  4 12:24 faillog
-rw-r--r--  1 root root     7354 Sep  4 12:31 fontconfig.log
drwxr-xr-x  2 _gvm _gvm      4096 Apr 23 10:32 gvm
```


/var/log

```
(kali㉿kali)-[/var/log]
```

```
$ ls -la
```

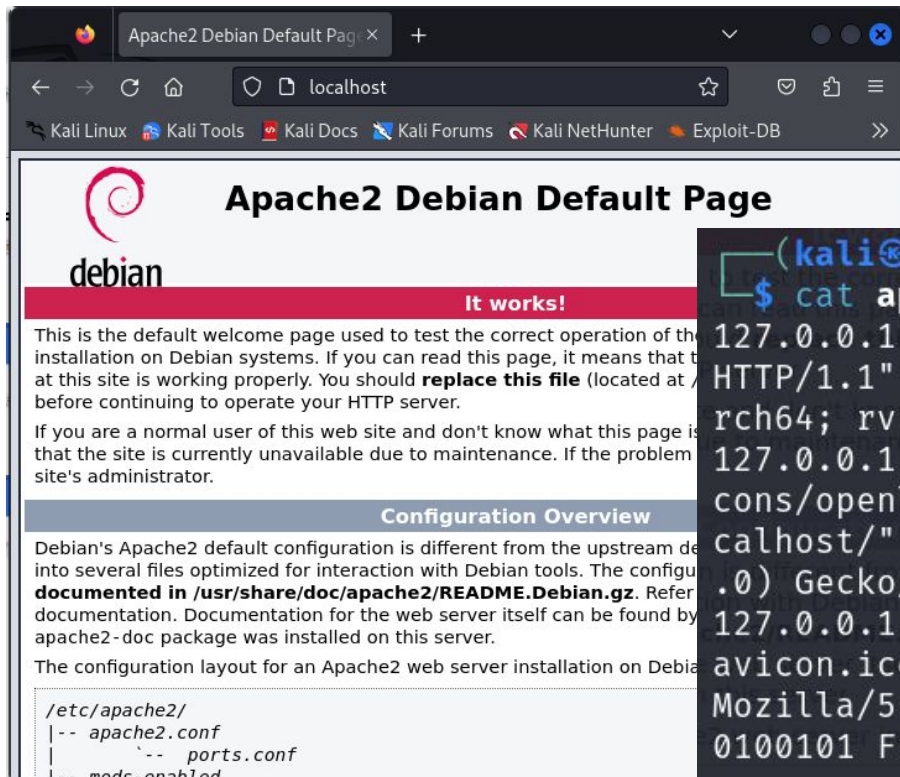
```
total 1596
drwxr-xr-x 21 root root 4096 Sep  7 12:34 .
drwxr-xr-x 12 root root 4096 Sep  4 19:34 ..
lrwxrwxrwx  1 root root   39 Sep  4 12:24 README → ../usr/s
hare/doc/systemd/README.logs
-rw-r--r--  1 root root 22869 Sep  7 12:35 Xorg.0.log
-rw-r--r--  1 root root 21855 Sep  7 12:35 Xorg.1.log
-rw-r--r--  1 root root 82365 Sep  4 12:29 alternatives.log
drwxr-xr-x  2 root adm 4096 Sep  4 12:28 apache2
drwxr-xr-x  2 root root 4096 Sep  4 12:55 apt
-rw-r--r--  1 root root 6531 Sep  4 19:34 boot.log
-rw-rw-r--  1 root utmp      0 Sep  4 12:24 btmp
-rw-r--r--  1 root root 1327536 Sep  4 12:55 dpkg.log
-rw-r--r--  1 root root      0 Sep  4 12:24 faillog
-rw-r--r--  1 root root 7354 Sep  4 12:31 fontconfig.log
drwxr-xr-x  2 _gvm _gvm 4096 Apr 23 10:32 gvm
```

```
(kali㉿kali)-[/var/log/apache2]
```

```
$ ls
```

```
access.log  error.log  other_vhosts_access.log
```

ตัวอย่าง access.log



```
(kali@kali)-[/var/log]
$ cat apache2/access.log
127.0.0.1 - - [12/Sep/2024:22:17:53 -0400] "GET / HTTP/1.1" 200 3380 "-" "Mozilla/5.0 (X11; Linux aarch64; rv:109.0) Gecko/20100101 Firefox/115.0"
127.0.0.1 - - [12/Sep/2024:22:17:53 -0400] "GET /icons/openlogo-75.png HTTP/1.1" 200 6040 "http://localhost/" "Mozilla/5.0 (X11; Linux aarch64; rv:109.0) Gecko/20100101 Firefox/115.0"
127.0.0.1 - - [12/Sep/2024:22:17:53 -0400] "GET /favicon.ico HTTP/1.1" 404 487 "http://localhost/" "Mozilla/5.0 (X11; Linux aarch64; rv:109.0) Gecko/20100101 Firefox/115.0"
```

Explain Web Server Log

192.168.1.2 - - [01/Sep/2024:12:05:30 +0000]
"POST /login HTTP/1.1" 302 512
"https://example.com/login" "Mozilla/5.0
(Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko)
Firefox/89.0"

Explain Web Server Log

192.168.1.2 - - → ที่อยู่ IP ของผู้ใช้ที่เข้าถึงเว็บไซต์

[01/Sep/2024:12:05:30 +0000] → วันที่และเวลาเมื่อคำขอถูกส่งไปที่เซิร์ฟเวอร์

"POST /login HTTP/1.1" → ข้อมูลเกี่ยวกับคำขอ HTTP ที่ถูกส่งมา (Request)

302 → สถานะการร้องขอที่ตอบกลับจากเซิร์ฟเวอร์ (Response Code)

512 → ขนาดของข้อมูลที่ส่งกลับจากเซิร์ฟเวอร์ในหน่วยไบต์ (Response Data)

"https://example.com/login" → เว็บไซต์ที่ผู้ใช้ต้องการเข้าถึง

"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Firefox/89.0" → User Agent: ข้อมูลเกี่ยวกับเบราว์เซอร์ที่ผู้ใช้งาน

ทำ Lab เกี่ยวกับ Log ไฟล์

Practice Time: Log Analysis

```
114.126.169.192 - - [08/Sep/2024:16:44:47 ] "GET /
profile HTTP/1.1" 500 364
"https://www.torres.info/"
"Mozilla/5.0 (X11; Linux
x86_64) AppleWebKit/533.2
(KHTML, like Gecko)
Chrome/48.0.811.0 Safari/
533.2"
216.192.152.55 - - [08/Sep/2024:16:44:47 ] "GET /
login HTTP/1.1" 500 440
"https://
www.richmond.com/"
"Mozilla/5.0 (iPhone; CPU
```

for_lab1_access.log

Log File - 30 KB

Information

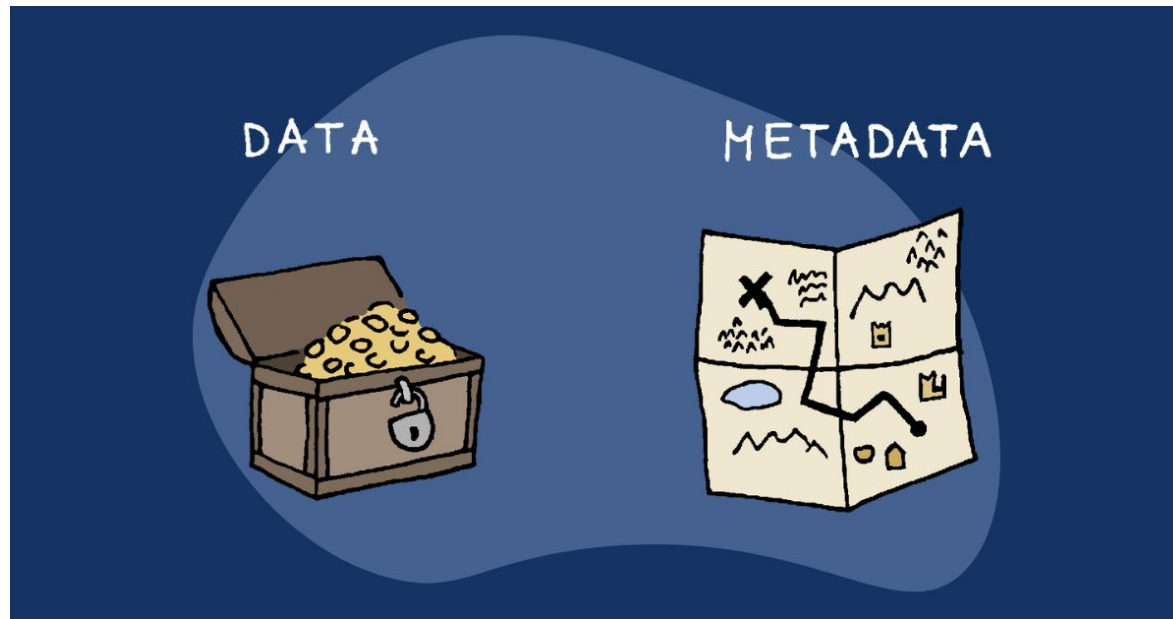
Find the Flag in for_lab1_access.log

for_lab1_access.log

Reveal Now Clear Reload Share

```
114.126.169.192 - - [08/Sep/2024:16:44:47 ] "GET /profile HTTP/1.1" 500 364 "https://www.torres.info/"
"Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/533.2 (KHTML, like Gecko) Chrome/48.0.811.0 Safari/533.2"
216.192.152.55 - - [08/Sep/2024:16:44:47 ] "GET /login HTTP/1.1" 500 440 "https://www.richmond.com/" "Mozilla/
5.0 (iPhone; CPU iPhone OS 16_7_6 like Mac OS X) AppleWebKit/535.1 (KHTML, like Gecko) FxiOS/12.0y1320.0
Mobile/26V247 Safari/535.1"
69.219.174.247 - - [08/Sep/2024:16:44:47 ] "DELETE /home HTTP/1.1" 200 1413 "http://www.richardson-
pineda.com/" "Mozilla/5.0 (Macintosh; U; Intel Mac OS X 10_10_7 rv:6.0; iw-IL) AppleWebKit/535.37.6 (KHTML,
like Gecko) Version/5.1 Safari/535.37.6"
54.177.214.133 - - [08/Sep/2024:16:44:47 ] "DELETE /login HTTP/1.1" 500 1284 "http://lewis.com/" "Mozilla/5.0
(iPod; U; CPU iPhone OS 4_2 like Mac OS X; apn-IN) AppleWebKit/534.13.3 (KHTML, like Gecko) Version/4.0.5
Mobile/8B119 Safari/6534.13.3"
8.178.79.227 - - [08/Sep/2024:16:44:47 ] "PUT /profile HTTP/1.1" 200 1451 "http://cordova.net/" "Mozilla/5.0
(Linux; Android 1.0) AppleWebKit/533.0 (KHTML, like Gecko) Chrome/60.0.894.0 Safari/533.0"
79.217.37.82 - - [08/Sep/2024:16:44:47 ] "DELETE /home HTTP/1.1" 500 1495 "http://wood.net/" "Mozilla/5.0
(X11; Linux x86_64; rv:1.9.7.20) Gecko/6080-04-11 06:23:58.862411 Firefox/3.6.9"
128.121.155.39 - - [08/Sep/2024:16:44:47 ] "POST /home HTTP/1.1" 404 1320 "http://potts.com/" "Opera/9.39.
(Windows NT 5.01; cmn-TW) Presto/2.9.170 Version/10.00"
52.195.39.104 - - [08/Sep/2024:16:44:47 ] "PUT /login HTTP/1.1" 200 962 "https://www.torres.net/" "Mozilla/5.0
(iPad; CPU iPad OS 4_3_5 like Mac OS X) AppleWebKit/532.2 (KHTML, like Gecko) FxiOS/18.2r0889.0 Mobile/400521
Safari/532.2"
66.67.75.185 - - [08/Sep/2024:16:44:47 ] "DELETE /login HTTP/1.1" 200 1791 "https://www.combs-young.biz/"
"Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/531.2 (KHTML, like Gecko) Chrome/25.0.816.0 Safari/531.2"
124.3.18.253 - - [08/Sep/2024:16:44:47 ] "DELETE /home HTTP/1.1" 404 1910 "https://www.johnston.com/" "Opera/
8.32.(X11; Linux i686; ia-FR) Presto/2.9.189 Version/10.00"
208.108.149.255 - - [08/Sep/2024:16:44:47 ] "DELETE /signup HTTP/1.1" 200 1435 "http://www.mcknight.com/"
"Mozilla/5.0 (Macintosh; PPC Mac OS X 10_5_4 rv:6.0; sid-ET) AppleWebKit/535.25.2 (KHTML, like Gecko) Version/
```

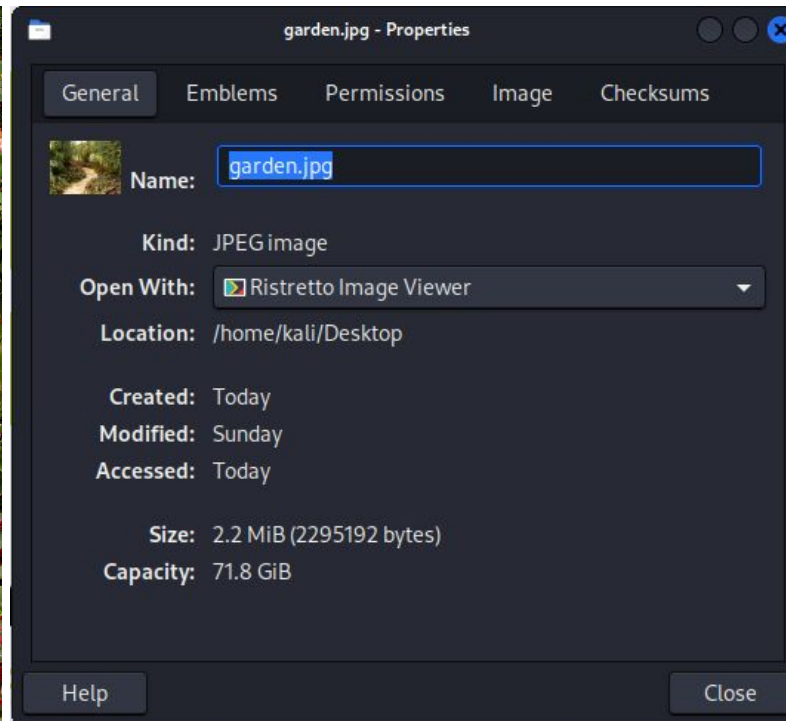
Metadata



<https://dataedo.com/blog/data-vs-metadata>

EXIF Data is metadata attached to photos which can include location, time, and device information.

Metadata



ที่มา: <https://picoctf.org>

เครื่องมือในการวิเคราะห์ Metadata

Offline Tool

- ExifTool
- Binwalk
- pdftinfo

Online Tool

- <https://gchq.github.io/CyberChef/>
- <https://exif.tools/>
- etc.

Forensics
Detect File Type
Scan for Embedded Files
Extract Files
YARA Rules
Remove EXIF
Extract EXIF
Extract RGBA
View Bit Plane
Randomize Colour Palette
Extract LSB
ELF Info

ExifTool

```
root@kali:~# exiftool -h
Syntax:  exiftool [OPTIONS] FILE

Consult the exiftool documentation for a full list of options.
```

← → ↺ 🏠 ⓘ exiftool.org

ExifTool by Phil Harvey

Read, Write and Edit Meta Information!

Also available --> [Utility to fix Nikon NEF images corrupted by Nikon software](#)

ตัวอย่างการใช้งาน ExifTool

```
(kali㉿kali)-[~/Desktop]
$ exiftool garden.jpg

ExifTool Version Number      : 12.76
File Name                    : garden.jpg
Directory                    : .
File Size                    : 2.3 MB
File Modification Date/Time   : 2024:09:04 13:06:45-04:00
File Access Date/Time        : 2024:09:09 13:31:31-04:00
File Inode Change Date/Time   : 2024:09:04 13:07:40-04:00
File Permissions              : -rw-r--r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : inches
X Resolution                  : 72
Y Resolution                  : 72
Profile CMM Type              : Linotronic

Profile CMM Type              : Linotronic
Profile Version               : 2.1.0
Profile Class                  : Display Device Profile
Color Space Data              : RGB
Profile Connection Space      : XYZ
Profile Date Time             : 1998:02:09 06:49:00
Profile File Signature        : acsp
Primary Platform              : Microsoft Corporation
CMM Flags                     : Not Embedded, Independent
Device Manufacturer          : Hewlett-Packard
Device Model                  : sRGB
Device Attributes             : Reflective, Glossy, Positive, Color
Rendering Intent              : Perceptual
Connection Space Illuminant   : 0.9642 1 0.82491
Profile Creator               : Hewlett-Packard
Profile ID                    : 0
Profile Copyright             : Copyright (c) 1998 Hewlett-Packard Company
```

ตัวอย่างคำสั่ง ExifTool

0) แยกข้อมูลออกจากไฟล์

```
$ exiftool a.jpg
```

1) เขียน Metadata ลงในไฟล์

```
$ exiftool -artist=me a.jpg
```

2) เขียน Metadatas Properties ลงในไฟล์

```
$ exiftool -artist="Phil Harvey" -copyright="2011 Phil Harvey" a.jpg
```

ดูเพิ่มเติมได้ที่: <https://exiftool.org/examples.html>

Practice Time: Information

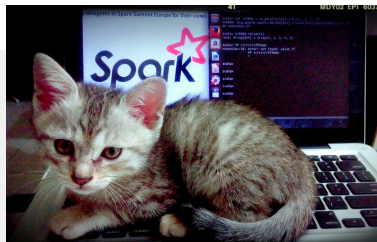
Find the Flag in for_lab2_cat.jpg



ตัวอย่างโจทย์จาก <https://picoctf.org>

Practice Time: Information

```
$ exiftool for_lab2_cat.jpg
ExifTool Version Number      : 12.76
File Name                    : cat.jpg
Directory                    : .
File Size                     : 878 kB
File Modification Date/Time   : 2024:09:04 15:38:20+00:00
File Access Date/Time        : 2024:09:04 15:41:40+00:00
File Inode Change Date/Time   : 2024:09:04 15:39:40+00:00
File Permissions              : -rw-r--r--
File Type                     : JPEG
File Type Extension           : jpg
MIME Type                     : image/jpeg
JFIF Version                  : 1.02
Resolution Unit                : None
X Resolution                   : 1
Y Resolution                   : 1
Current IPTC Digest           : 7a78f3d9cfb1ce42ab5a3aa30573d617
Copyright Notice              : PicoCTF
Application Record Version    : 4
XMP Toolkit                   : Image::ExifTool 10.80
License                       : picoCTF{ME74D47A_HIDD3N_3b9209a2}
Rights                        : PicoCTF
```



ทำไมถึงรู้



นี่แหละความลับของผมกับต้น



ผมใช้ Exiftool

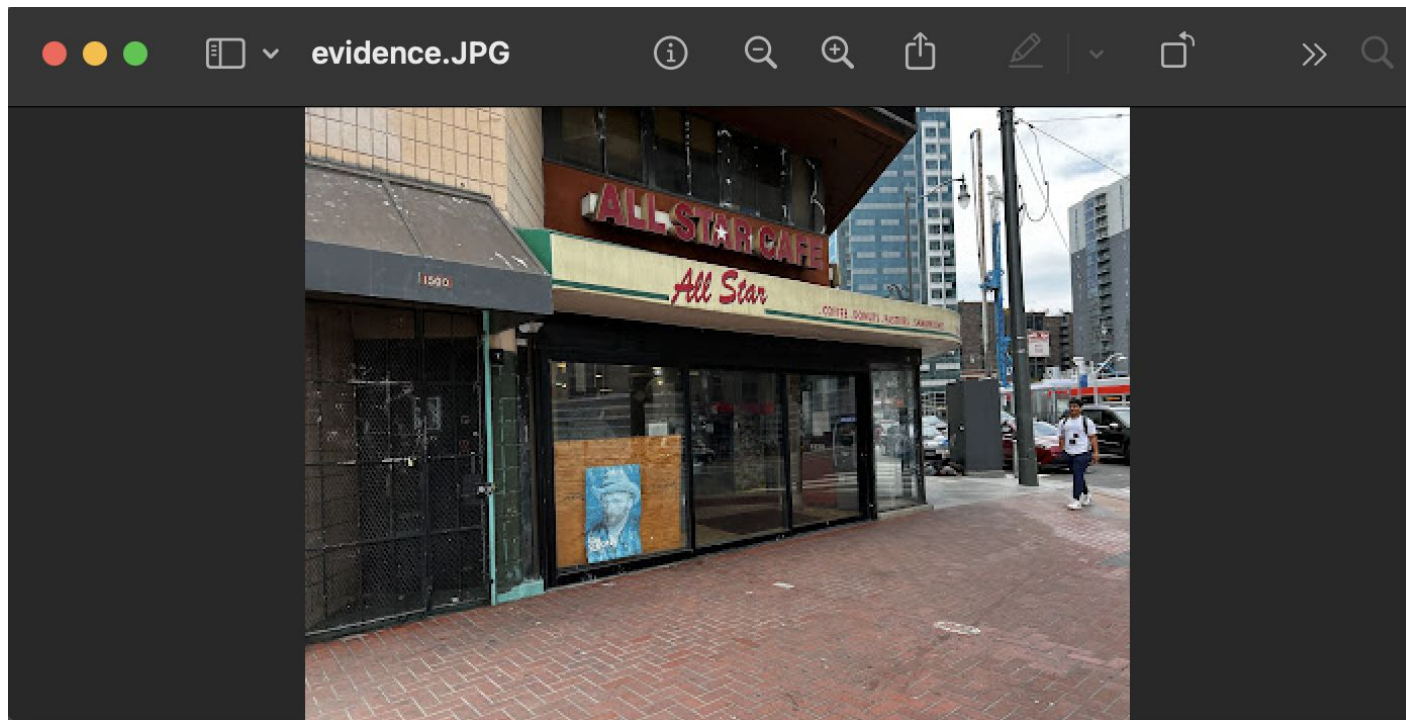
Forensics - โจทย์สมมติ

สถานการณ์:

หน่วยสืบสวนกำลังตามหาผู้ต้องสงสัยในการโจรกรรมเงินจำนวน
มหาศาล ผู้ต้องสงสัยหนีไปพร้อมกับเงินและได้หลบซ่อนตัว โดยใช้
รถยนต์เป็นพาหนะ ข้อมูลเพียงอย่างเดียวที่ตำรวจมีคือรูปถ่ายหนึ่งรูป
ที่ถ่ายจากกล้องมือถือในวันเกิดเหตุ อย่างไรก็ตาม พิกัด GPS จากรูป
นั้นอาจเป็นเบาะแสสำคัญที่จะนำไปสู่สถานที่ที่ผู้ต้องสงสัยกำลังใช้รถ
ยี่ห้อ Toyota ในการหลบหนีแต่ไม่ทราบรุ่นของรถ

Forensics - โจทย์สมมติ

หลักฐานจากโจทย์ evidence.jpg



Forensics - โจทย์สมมติ

ใช้ exiftool เพื่อดู Metadata ของหลักฐาน

```
$ exiftool evidence.JPG
ExifTool Version Number      : 12.76
File Name                    : evidence.JPG
Directory                    : .
File Size                    : 59 kB
File Modification Date/Time   : 2024:09:13 15:15:30-04:00
File Access Date/Time        : 2024:09:13 15:15:30-04:00
File Inode Change Date/Time   : 2024:09:13 15:15:30-04:00
File Permissions              : -rw-r--r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : None
```

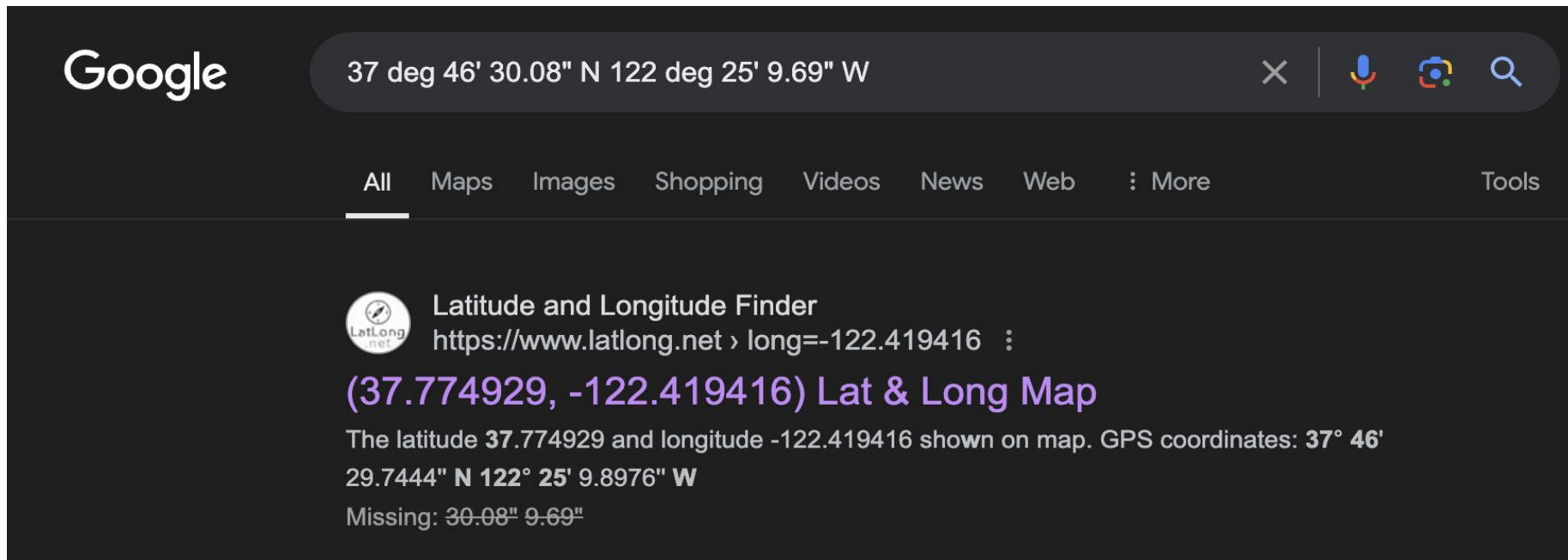

Forensics - โจทย์สมมติ

ใช้ exiftool เพื่อดู Metadata ของหลักฐาน

```
GPS Version ID           : 2.3.0.0
GPS Latitude Ref         : North
GPS Longitude Ref        : West
Image Width              : 408
Image Height             : 306
Encoding Process         : Baseline DCT, Huffman coding
Bits Per Sample          : 8
Color Components         : 3
Y Cb Cr Sub Sampling     : YCbCr4:4:4 (1 1)
Image Size               : 408x306
Megapixels               : 0.125
GPS Latitude             : 37 deg 46' 30.08" N
GPS Longitude            : 122 deg 25' 9.69" W
GPS Position             : 37 deg 46' 30.08" N, 122 deg 25' 9.69" W
```

Forensics - โจทย์สมมติ

ค้นหา พิกัด GPS บน Google Search



The screenshot shows a Google search interface with a dark theme. The search bar contains the coordinates "37 deg 46' 30.08" N 122 deg 25' 9.69" W". Below the search bar, the "All" tab is selected. The search results display a link to "Latitude and Longitude Finder" with the URL "https://www.latlong.net › long=-122.419416". The title of the result is "(37.774929, -122.419416) Lat & Long Map" in purple. The description states: "The latitude 37.774929 and longitude -122.419416 shown on map. GPS coordinates: 37° 46' 29.7444" N 122° 25' 9.8976" W". Below this, it says "Missing: 30.08" 9.69"

Google

37 deg 46' 30.08" N 122 deg 25' 9.69" W

All Maps Images Shopping Videos News Web : More Tools

 Latitude and Longitude Finder
<https://www.latlong.net › long=-122.419416> :

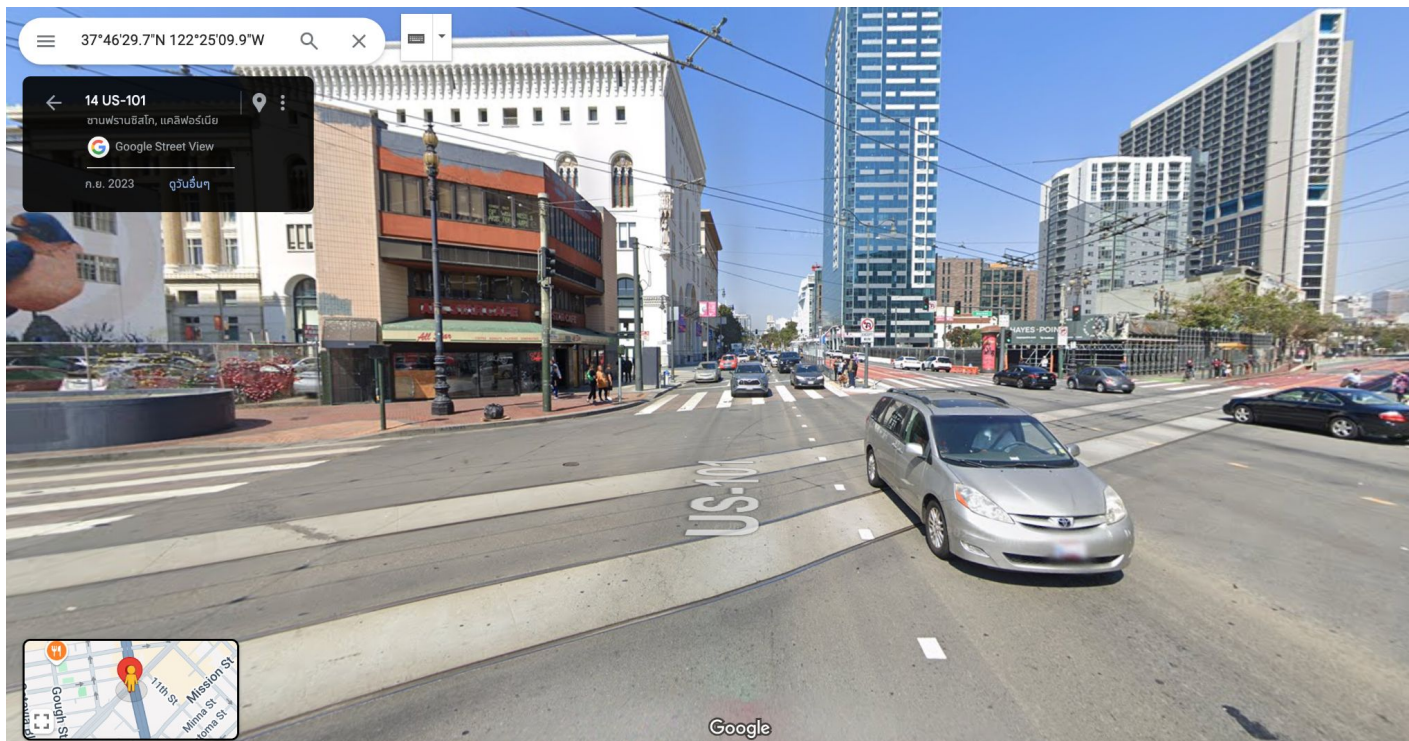
(37.774929, -122.419416) Lat & Long Map

The latitude 37.774929 and longitude -122.419416 shown on map. GPS coordinates: 37° 46' 29.7444" N 122° 25' 9.8976" W

Missing: 30.08" 9.69"

Forensics - โจทย์สมมติ

ค้นหา พิกัด GPS บน Google Map



Forensics - โจทย์สมมติ

ใช้ Google Image Search

Google

Find image



Wikimedia

File:Toyota-Sienna-CE.jpg - Wikimedia...



Related searches

2010 Toyota Sienna

Toyota Sienna



Wikimedia

File:Toyota Sienna LE (6222546663).jpg - ...



Wikimedia

File:Toyota-Sienna- - Wikimedia Comm



The New York Times
Wheelies: The Toyota Trouble Edition - The...



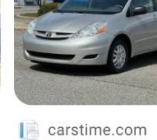
YouTube

Toyota Sienna Review (2006) - AutosForSale...



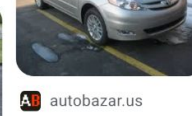
Wikimedia

File:Toyota-Sienna-CE.jpg - Wikimedia...



carstime.com

Toyota Sienna For Sale in Linden, NJ - Car...

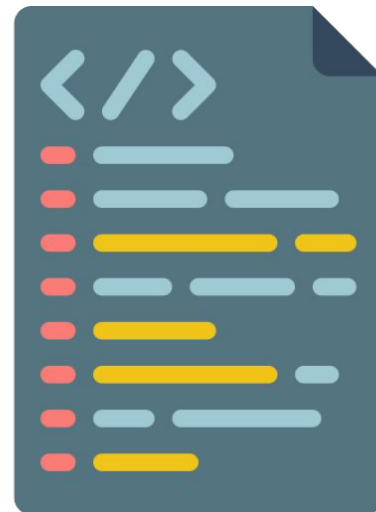
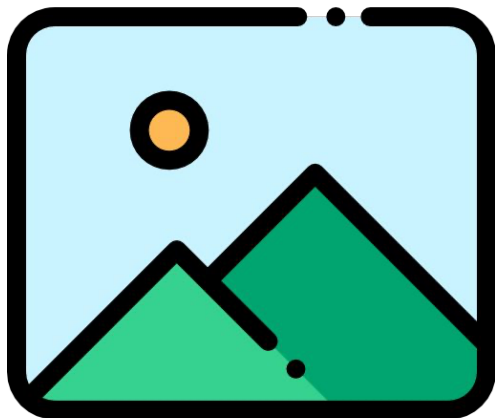


autobazar.us





File Format



File Format



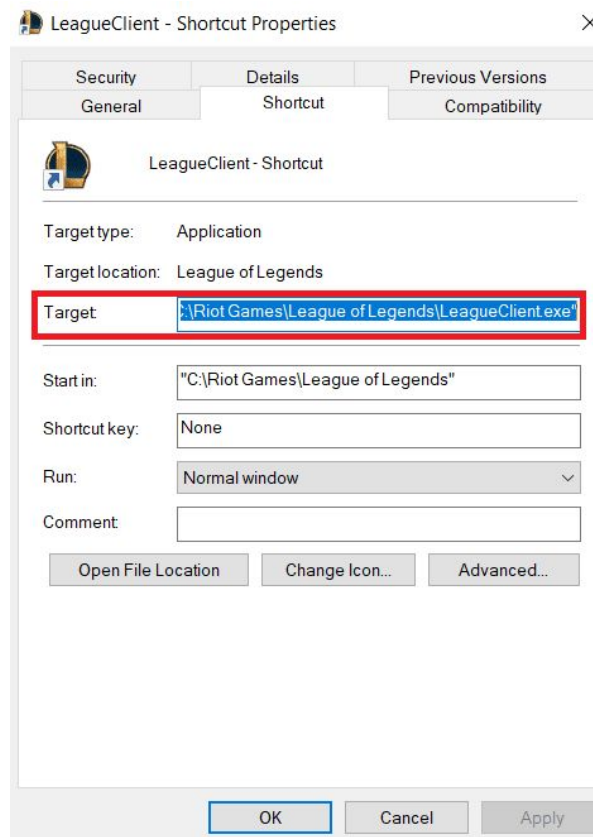
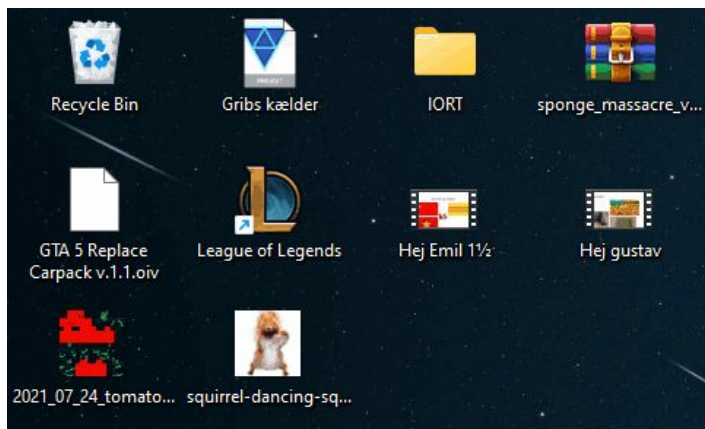
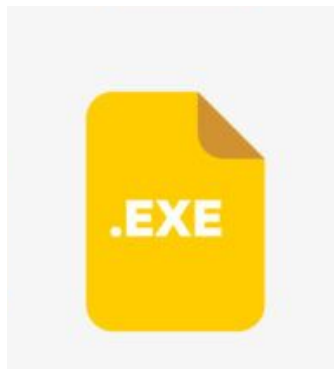
ที่มา: <https://www.thebrandcollective.co.za>

File Format



```
▶ note.txt
1 Hello, world
2
3 This is my first text.
4
5
```


File Format

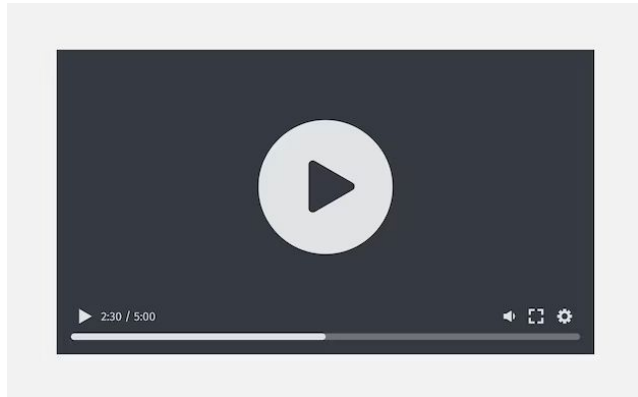


File Format



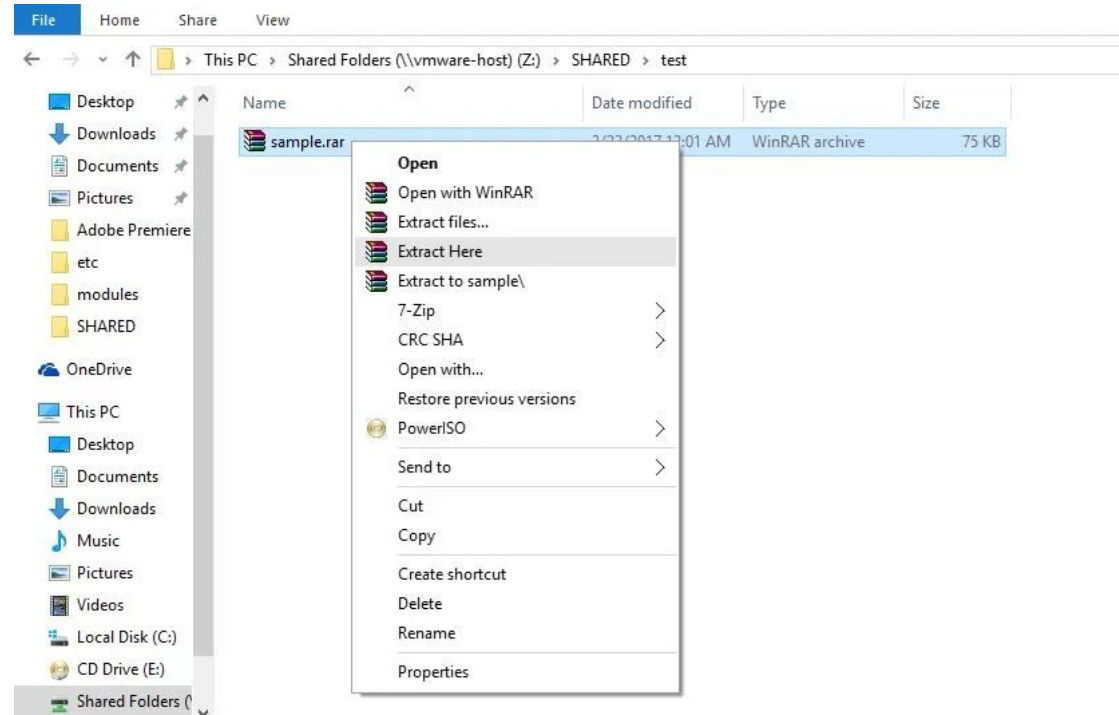
garden.jpg

File Format

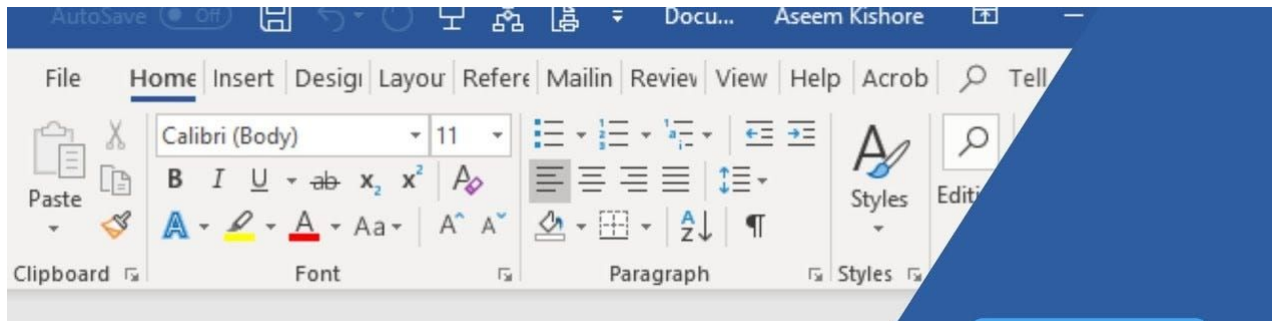
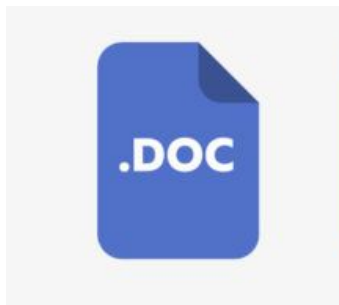


ที่มา: <https://www.freepik.com>

File Format



File Format



MS - Word



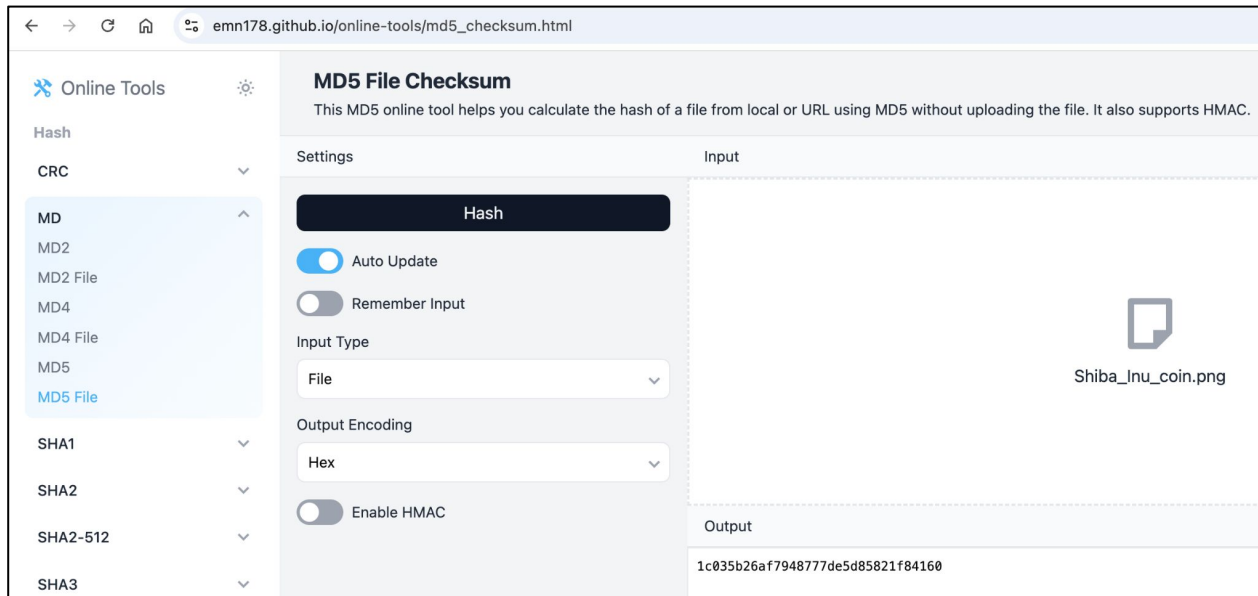
เครื่องมือที่ใช้ในการวิเคราะห์ File Format

Offline Tool

- Hex Editor
- ExifTool
- File
- Checksum
 - md5sum
 - sha1sum
 - sha256sum
 - etc.

Online Tool

- <https://gchq.github.io/CyberChef/>
- https://emn178.github.io/online-tools/md5_checksum.html



The screenshot shows the 'MD5 File Checksum' web application. The left sidebar lists various hash algorithms, with 'MD' selected and 'MD5 File' highlighted. The main area has a 'Settings' panel on the left and an 'Input' area on the right. The 'Settings' panel includes a 'Hash' dropdown set to 'MD5', an 'Auto Update' toggle (checked), a 'Remember Input' toggle (unchecked), an 'Input Type' dropdown set to 'File', an 'Output Encoding' dropdown set to 'Hex', and an 'Enable HMAC' toggle (unchecked). The 'Input' area contains a large text box with a file icon and the text 'Shiba_Inu_coin.png'. Below the input area, the 'Output' section displays the MD5 hash: '1c035b26af7948777de5d85821f84160'.

File command

```
(kali㉿kali)-[~/Desktop]
```

```
$ file cat.jpg
```

```
cat.jpg: JPEG image data, JFIF standard 1.02, aspect ratio, density  
1×1, segment length 16, Exif Standard: [TIFF image data, big-endian,  
direntries=5, xresolution=74, yresolution=82, resolutionunit=1, cop  
yright=Hello, World], baseline, precision 8, 2560×1598, components 3
```

```
(kali㉿kali)-[~/Desktop]
```

```
$ file admin.php
```

```
admin.php: PHP script, ASCII text
```


File command

```
(kali㉿kali)-[~/Desktop]
$ file info.pdf
info.pdf: PDF document, version 1.3, 1 page(s)
```

```
(kali㉿kali)-[~/Desktop]
$ file archive.zip
archive.zip: Zip archive data, at least v2.0 to extract, compression
method=deflate
```

File Signature

Hex signature	ISO 8859-1	Offset	Extension	Description
23 21	#!	0		Script or data to be passed to the program following the shebang (#!) ^[1]
02 00 5a 57 52 54 00 00 00 00 00 00 00 00 00 00	STX NUL ZWRT NUL NUL NUL NUL NUL NUL NUL NUL	0	cwk	Claris Works word processing doc
00 00 02 00 06 04 06 00 08 00 00 00 00 00	NUL NUL STX NUL ACK EOT ACK NUL BS NUL NUL NUL NUL	0	wk1	Lotus 1-2-3 spreadsheet (v1) file
25 50 44 46 2D	%PDF-	0	pdf	PDF document ^[34]
30 26 B2 75 8E 66 CF 11 A6 D9 00 AA 00 62 CE 6C	0&²užfİ DCI !Ü NUL a NUL bÎl	0	asf wma wmv	Advanced Systems Format ^[35]

https://en.wikipedia.org/wiki/List_of_file_signatures

Hex Editor

\$ hexeditor garden

File: garden	ASCII Offset: 0-00000000 / 0-00230597 (x00)																
00000000	FF	D8	FF	E0	00	10	4A	46	49	46	00	01	01	01	00	48	... JFIF.....H
00000010	00	48	00	00	FF	E2	0C	58	49	43	43	5F	50	52	4F	46	XICC_PROF
00000020	49	4C	45	00	01	01	00	00	0C	48	4C	69	6E	6F	02	10	HLino..
00000030	00	00	6D	6E	74	72	52	47	42	20	58	59	5A	20	07	CE	ntrRGB XYZ ..
00000040	00	02	00	09	00	06	00	31	00	00	61	63	73	70	4D	53	I....1..acspMS
00000050	46	54	00	00	00	00	49	45	43	20	73	52	47	42	00	00	... IEC sRGB ..
00000060	00	00	00	00	00	00	00	00	00	00	00	00	F6	D6	00	01	
00000070	00	00	00	00	D3	2D	48	50	20	20	00	00	00	00	00	00	..-HP
00000080	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000090	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
000000A0	00	00	00	00	00	00	00	00	00	11	63	70	72	74	00	00	cprt ..
000000B0	01	50	00	00	00	33	64	65	73	63	00	00	01	84	00	00	..3desc.....
000000C0	00	6C	77	74	70	74	00	00	01	F0	00	00	00	14	62	6B	tpt.....bk
000000D0	70	74	00	00	02	04	00	00	00	14	72	58	59	5A	00	00	rXYZ ..
000000E0	02	18	00	00	00	14	67	58	59	5A	00	00	02	2C	00	00	...gXYZ ... , ..
000000F0	00	14	62	58	59	5A	00	00	02	40	00	00	00	14	64	6D	XYZ ... @....dm
00000100	6E	64	00	00	02	54	00	00	00	70	64	6D	64	64	00	00	..T ... pdmdd ..
00000110	02	C4	00	00	00	88	76	75	65	64	00	00	03	4C	00	00	...vued ... L..
00000120	00	86	76	69	65	77	00	00	03	D4	00	00	00	24	6C	75	iew.....\$lu

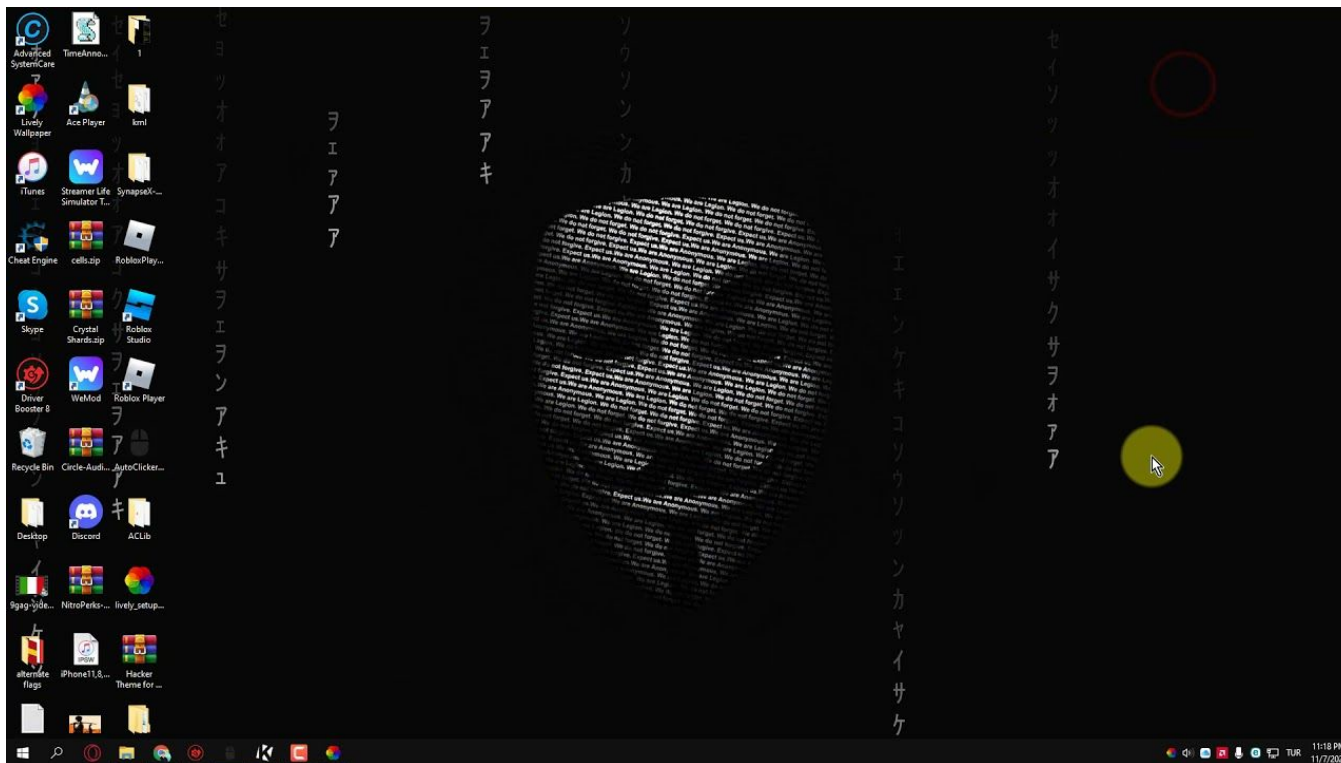
File Signature

FF D8 FF DB	ÿøÿÙ			
FF D8 FF E0 00 10 4A 46 49 46 00 01	ÿøÿà NUL DLE JFIF NUL SOH	0	jpg jpeg	JPEG raw or in the JFIF or Exif file format ^[16]
FF D8 FF EE	ÿøÿî			
FF D8 FF E1 ?? ?? 45 78 69 66 00 00	ÿøÿá??Exif NUL NUL			
FF D8 FF E0	ÿøÿà	0	jpg	JPEG raw or in the JFIF or Exif file format ^[16]

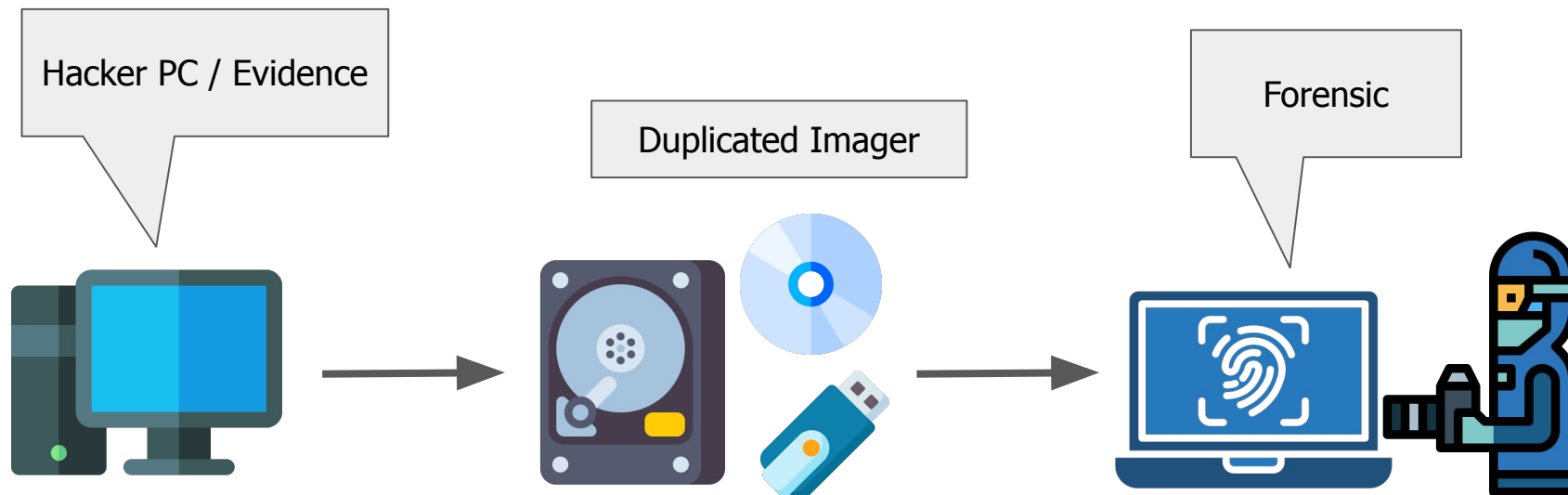
ทำไม File Format และ File Signature ถึงสำคัญ?

- ระบุประเภทของไฟล์
- ตรวจสอบความถูกต้องของไฟล์
- กู้คืนข้อมูล
- ตรวจสอบความปลอดภัย

Disk Imaging



Disk Imaging



ตัวอย่าง Autopsy

008 - Autopsy 4.15.0

Case View Tools Window Help

+ Add Data Source Images/Videos Communications Geolocation Timeline File Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing File System 261 Results

Table Thumbnail

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Keywords
nb-NO			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		ur
zh-TW			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		ur
connection			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		ur
tool			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		ur
System			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		ur
System			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		ur
FDs			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		ur
Jetco			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		ur
Micro			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		ur
Spide			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		ur
System			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		ur

Properties
View File in Directory
View in New Window
Open in External Viewer Ctrl+E
View File in Timeline...
Extract File(s)
Export selected rows to CSV
Add File Tag
Remove File Tag
Add File to Hash Set (Empty File)

Hex Text Results Annotations Other Occurrences

Name File System
Type application/octet-stream
Size 0
File Name Allocation Unallocated
Metadata Allocation
Modified 0000-00-00 00:00:00
Accessed 0000-00-00 00:00:00
Created 0000-00-00 00:00:00
Changed 0000-00-00 00:00:00
MD5 d41d8cd98f00b204e9800998ecf8427e
Hash Lookup Results UNKNOWN

Data Sources

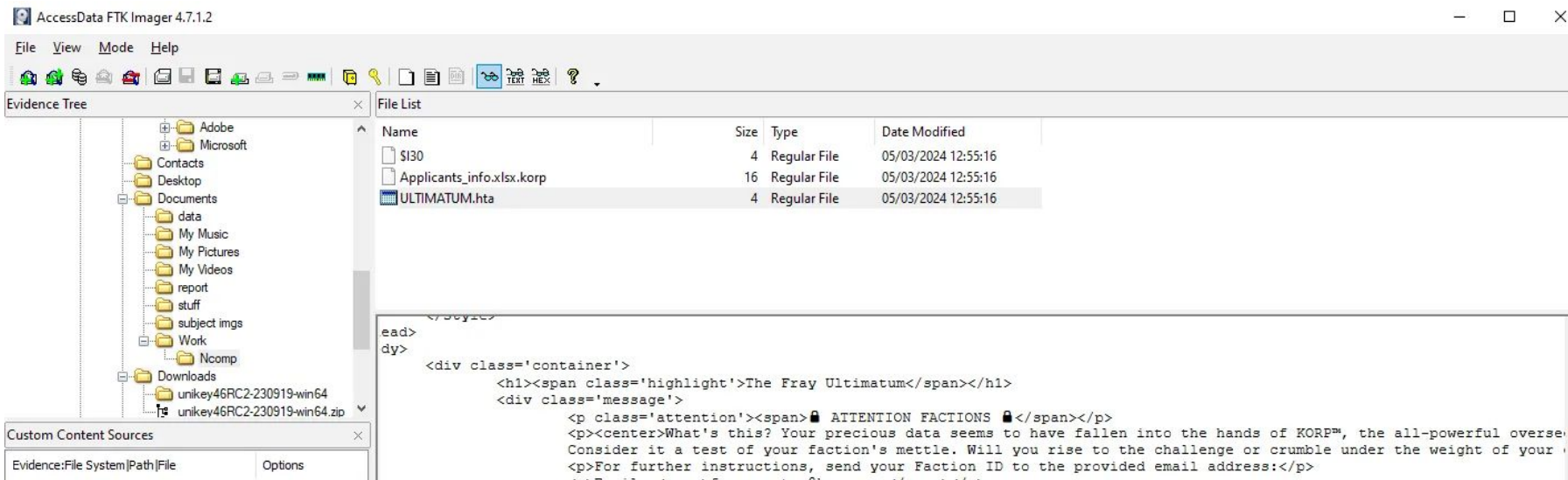
- Mantooth.E01
 - vol1 (Unallocated: 0-62)
 - vol2 (NTFS / exFAT (0x07): 63-224909)
 - vol3 (DOS FAT12 (0x01): 224910-240974)
 - vol4 (Unallocated: 240975-250878)

Views

- File Types
 - By Extension
 - By MIME Type
- Deleted Files
 - File System (261)
 - All (279)
- MB File Size
 - MB 50 - 200MB (2)
 - MB 200MB - 1GB (0)
 - MB 1GB+ (0)

Results

- Extracted Content
 - EXIF Metadata (8)
 - Encryption Detected (2)
 - Extension Mismatch Detected (9)
 - Installed Programs (40)
 - Operating System Information (2)
 - Operating System User Account (8)
 - Recent Documents (109)
 - Recycle Bin (9)
 - Remote Drive (1)
 - Shell Bags (110)
 - USB Device Attached (30)
 - User Content Suspected (8)
 - Web Cookies (50)
 - Web History (374)
 - Web Search (130)
- Keyword Hits
 - Single Literal Keyword Search (0)
 - Single Regular Expression Search (0)



File hash (Checksum)

DOWNLOAD ETTERCAP SOURCE CODE

The latest Ettercap release is: 0.8.3.1-Bertillon
Release date: August 1, 2020

Click to Download the version with bundled libraries



[ettercap-0.8.3.1.tar.gz](#)

Sha1: 9185c59883c8d128b852233c9aa0ae46f772d170

1. ตรวจสอบความสมบูรณ์ของไฟล์

2. ช่วยในการยืนยันว่าไฟล์ที่ดาวน์โหลดมาจากแหล่ง
ที่มาที่เชื่อถือได้

PREVIOUS VERSIONS

Release Date	FileName	Version	Sha-1 Hash
Jul 1, 2019	ettercap-0.8.3.tar.gz	0.8.3-Bertillon	1db39315b2b2b574dc1eb3f7ae72871ad2391e2f
Jul 1, 2019	ettercap-0.8.3.tar.gz (source only)	0.8.3-Bertillon	90f2a397966d721868a1fb49c5ecaba0f0af72d0
Mar 2015	14, ettercap-0.8.2.tar.gz	0.8.2-Ferri	7e528632ca01c5977da1a0af56a5e05fbc383832
Mar 2015	14, ettercap-0.8.2.tar.gz (source only)	0.8.2-Ferri	04f50925a5f3b2555371075a048ab7bfe3892976
Oct 2014	16, ettercap-0.8.1.tar.gz	0.8.1-Lombroso	66362ce69cd9b82b9eb8ea6a52048700704a7d9b
Oct 2014	16, ettercap-0.8.1.tar.gz (source only)	0.8.1-Lombroso	1179923d94954cd6e00117c3492c4ca3991bc401
Sep 2013	21, ettercap-0.8.0.tar.gz	0.8.0-Lacassagne	008fca94bbd67b578699300eb321766cd41fbfff
Mar 2013	26, ettercap-0.7.6.tar.gz	0.7.6-Locard	55818952a8c28beb1b650f3ccc9600a2d784a18f

3. การป้องกันการปลอมแปลง

4. ใช้ในการตรวจจับข้อผิดพลาดที่อาจ

เกิดขึ้นระหว่างการถ่ายโอนข้อมูล

Forensic Lab 3 : Verify



ตัวอย่างโจทย์จาก
<https://picoctf.org>

```
(kali@kali)-[~/Desktop]
$ tree home
home
├── ctf-player
│   └── drop-in
│       ├── checksum.txt
│       ├── decrypt.sh
│       └── files
│           ├── 0agQiFLS
│           ├── 0pEkV2ds
│           ├── 0wWA41ot
│           ├── 0yVzp2am
│           ├── 12GUEFi0
│           ├── 12R70dbh
│           ├── 1EQhRC4i
│           ├── 1FjaHS3F
│           ├── 1cYEYb6L
│           ├── 1iXLQGXR
│           ├── 2h0QXHZC
│           └── 2nsMaCTj
```

```

├── wr59gSPm
├── wtPMewZq
├── wtq06VT7
├── wx3RfP7B
├── xzvNiQwK
├── y0EWonka
├── yajqgzPt
├── ypsNLNOA
├── ytd5LOm1
├── yzqsPNuQ
├── zH4qslwZ
├── zM5KAlbJ
├── zMQ1nXew
├── zWeRABcB
├── zZe9EIdH
└── zhBiEB8c

4 directories, 303 files
```

Identification

?

D
A
T
A

```
(kali@kali)-[~/Desktop/home/ctf-player/drop-in]
$ cat checksum.txt
3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4

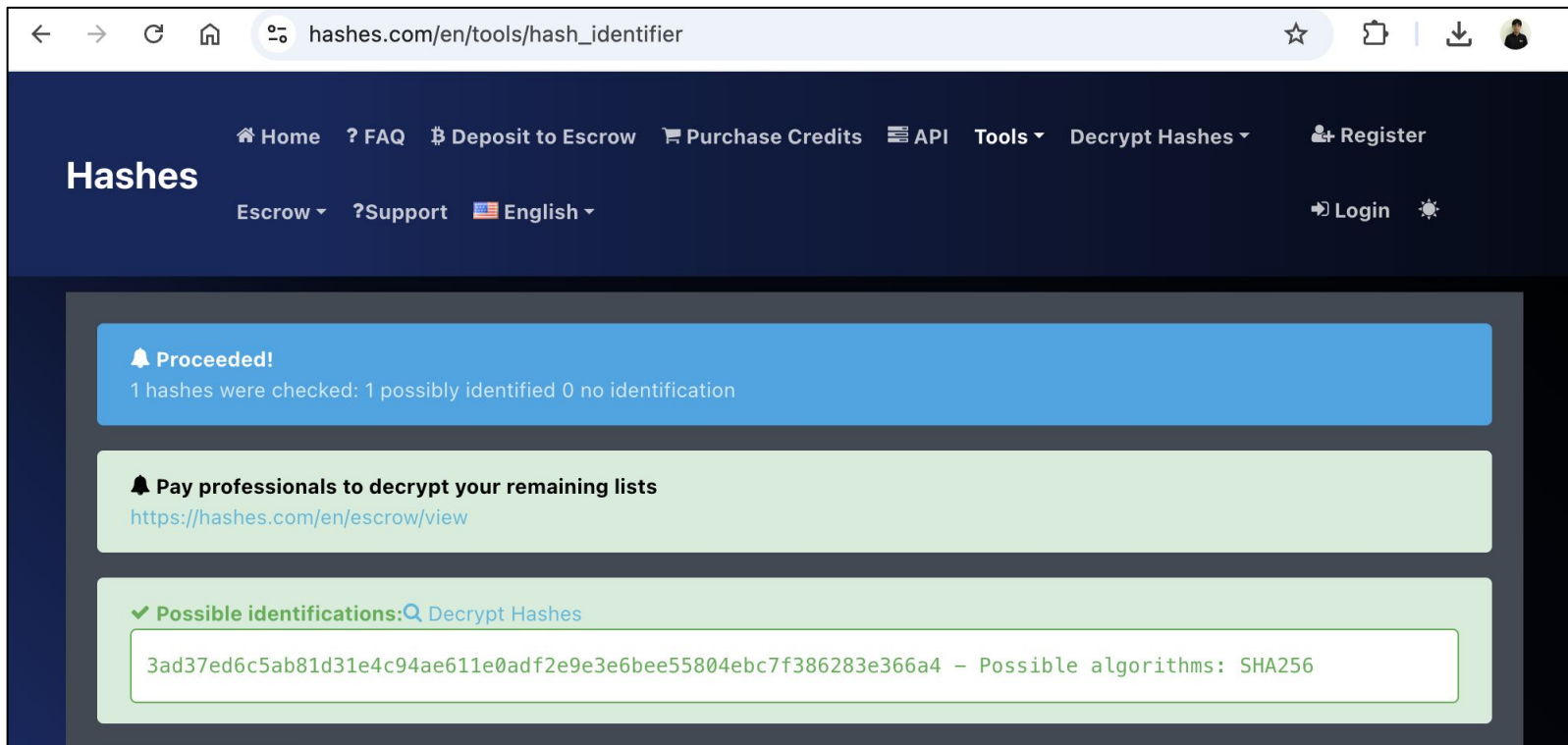
(kali@kali)-[~/Desktop/home/ctf-player/drop-in]
$ cat files/0agQiFLS
RliF1vdIPayu97ed5JRBhtZkNAuMHAEo1lthk9ky87Z6qdpqNhHUQEul2zxx9zY

(kali@kali)-[~/Desktop/home/ctf-player/drop-in]
$ cat files/0pEkV2ds
KRhaBTB9ByUUwhhLbbHvOybGIwo07QXHLRKgDacYYpGiisJ5mKUz623IMNxN6HJ

(kali@kali)-[~/Desktop/home/ctf-player/drop-in]
$ cat files/0wWA41ot
6f28qnMeSfYMP6Sq0XVshA99u4obPxptpqkBXXamzmOfhRLHdGKoeqKJIM6otSl
```

Hash Identifier

https://hashes.com/en/tools/hash_identifier



The screenshot shows the Hashes.com website interface. The browser address bar displays the URL https://hashes.com/en/tools/hash_identifier. The website has a dark blue header with navigation links: Home, FAQ, Deposit to Escrow, Purchase Credits, API, Tools, Decrypt Hashes, Register, Escrow, Support, and English. The main content area features a blue notification box stating "Proceeded! 1 hashes were checked: 1 possibly identified 0 no identification". Below this is a green box with a message: "Pay professionals to decrypt your remaining lists" with a link to <https://hashes.com/en/escrow/view>. At the bottom, another green box shows a "Possible identification" for the hash `3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4` with the algorithm identified as SHA256.

Hashes

Home ? FAQ Deposit to Escrow Purchase Credits API Tools Decrypt Hashes Register

Escrow ?Support English

Login

Proceeded!
1 hashes were checked: 1 possibly identified 0 no identification

Pay professionals to decrypt your remaining lists
<https://hashes.com/en/escrow/view>

Possible identifications: [Decrypt Hashes](#)

3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4 – Possible algorithms: SHA256

hashid

```
(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
$ hashid "3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4"
Analyzing '3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4'
[+] Snefru-256
[+] SHA-256
[+] RIPEMD-256
[+] Haval-256
[+] GOST R 34.11-94
[+] GOST CryptoPro S-Box
[+] SHA3-256
[+] Skein-256
[+] Skein-512(256)
```


sha256sum

```
(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
```

```
$ sha256sum files/0agQiFLS
```

```
885ac35e179e1b2746dcd163d980938090168c594dd7567bf1d60bdd11557df1 files/0agQiFLS
```

```
(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
```

```
$ ls files/
```

0agQiFLS	8cSetvuU	FmmWrIZ7	MwfNVxMB	SXltVhZT	XrAdESZl	covwBpER	kLFFhUci	t3YK8diU
0pEkV2ds	8sqe8FVs	GhWI4eJh	MzdMfZHa	SbFIvoSj	XrvFYdDg	cxKJcozU	l4lFl66y	tLG9HM3a
0wWA41ot	92q4JPFx	GkM1UPTw	NFea6BFS	SkQZlbB2	Xw38pYK0	dINee6RV	lUsUQJ4B	uJzNSw96
0yVzp2am	9EMX68VB	GoGhbQto	NgY5gymg	TBQytfqs	Xxuckerf	dS2gaUE4	m2KQa1Hp	uQyUDnOZ
12GUEFi0	9nLUSB5k	GodaoG3e	NopiZGwa	TSJCXwIT	Y4u4wEGY	dVJ9IeAT	m5c0AhxS	v6LGqmwI
12R70dbh	A0aXQwRy	H3rLRpMi	O2eRM15N	TTjL07LQ	YCf9VpOR	dZVn0thw	mVEKZ3oW	v8sVJpVd
1EQhRC4i	AEjqj8u	HG10KGnm	OVCZKr6X	ThekmVcy	YG1pCKDt	dkV6p1DF	mcjegRRr	vCUbox39
1FjaHS3F	AeCM4Vvt	HPGHI002	OX3IlkB9	TqHFzH54	YPNiaCgG	dtc6oz6G	n05ZtlwX	vTgToTLG
1cYeyb6L	AhVRy5sU	HRSTilo9	OYpH5Rfs	Tui6wJfr	Ydulshwq	e018b574	n7C2bpPk	vfn94Ek3
1iXLQGXR	Aqg5GrWn	HchfFzCW	Opx3E3FO	TwspwefZ	YdxjMT1r	e7ir0vB1	nDj0INiw	vnr7vUto

sha256sum

```
(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
```

```
$ sha256sum files/*
```

```
885ac35e179e1b2746dcd163d980938090168c594dd7567bf1d60bdd11557df1 files/0agQiFLS
d3cdf8f8222104727892c6c5f306efcce1e1e53e1510b575dabf838e5c17619b files/0pEkV2ds
a67c3339a9661ea181bf073e543b47580011bd210b3306497a4ae8c9a2124bd7 files/0wWA41ot
2940827966c3d8b31f87ab008d471dffbf0f80c918a669eb34d00909adaf4d406 files/0yVzp2am
645e24226aa77f498951e38713e103659fb7bb41ceccf7cc6268c226ad03d4b9 files/1EQhRC4i
441db2a78012080ec0c3b80d2b5b6d66720297b65db195ead59ea646a0674c6e files/1FjaHS3F
5170556dbf37f1aab8cb439c7fd8f3e56f8e4aa41ee2e86379ef4190f7623b6f files/1cYEBb6L
bf5e0d888b16e034331f90067b41834ae50673d9dcffa4c76ee4416836cbe919 files/1iXLQGXR
8313a303c94d4fd615073aedded65df0cdee87d5fd074c25c847b064291ff4b3 files/2h0QXHZC
d63d4ee0d986694f4df69960d5c54c8ea8c3822dd5b5be1065406bb9c03b8a6c files/2nsMaCTj
16bda5067a52a32595ac140466be993b585c75b0557e012ec874ac813fdd73fd files/2zpsEiQJ
99c46603b95fcb29d95bc71106fadb95df93182e119097522d8d9efc823b4ef0 files/3BrLDAbO
807309a496177fd97eae436d0158e21cc5f15ee43adc6a0275015e8785c7e4e1 files/3PmKbHhH
```

Step to Verify

```
(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
$ sha256sum files/* > checksum-all-file.txt
```

```
(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
$ ls
checksum-all-file.txt  checksum.txt  decrypt.sh  files
```

```
(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
$ cat checksum.txt
3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4
```

```
(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
$ cat checksum-all-file.txt | grep "3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7
86283e366a4"
3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4 files/e018b574
```

Step to Verify

```
(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
└─$ ls
checksum-all-file.txt  checksum.txt  decrypt.sh  files

(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
└─$ cp files/e018b574 .

(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
└─$ ls
checksum-all-file.txt  checksum.txt  decrypt.sh  e018b574  files

(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
└─$ ./decrypt.sh e018b574
Error: 'e018b574' is not a valid file. Look inside the 'files' folder with 'ls -R!'

(kali㉿kali)-[~/Desktop/home/ctf-player/drop-in]
└─$ cat e018b574
Salted__m♦uI♦♦♦♦8Ⓒ♦♦♦♦;♦Z♦q♦E♦+L♦♦♦♦♦G(%Al6q?k♦♦♦♦♦♦♦♦$♦qE♦
```


Step to Verify

```
# Check if the provided argument is a file and not a folder
if [ ! -f "/home/ctf-player/drop-in/$file_name" ]; then
    echo "Error: '$file_name' is not a valid file. Look inside the 'files' folder with 'ls -R!'"
    exit 1
fi

# If there's an error reading the file, print an error message
if ! openssl enc -d -aes-256-cbc -pbkdf2 -iter 100000 -salt -in "/home/ctf-player/drop-in/$file_name" -k picoCTF; then
    echo "Error: Failed to decrypt '$file_name'. This flag is fake! Keep looking!"
fi

(kali@kali)-[~/Desktop/home/ctf-player/drop-in]
$

(kali@kali)-[~/Desktop/home/ctf-player/drop-in]
$ openssl enc -d -aes-256-cbc -pbkdf2 -iter 100000 -salt -in e018b574 -k picoCTF
picoCTF{trust_but_verify_e018b574}
```

Thank you !!

เกม !!!!

Binwalk



```
(kali@kali)-[~/Desktop]
$ binwalk dolls.jpg
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	PNG image, 594 x 1104, 8-bit/color RGBA, non-interlaced
3226	0xC9A	TIFF image data, big-endian, offset of first image directory: 8
272492	0x4286C	Zip archive data, at least v2.0 to extract, compressed size: 378954, uncompressed size: 383938, name: base_images/2_c.jpg
651612	0x9F15C	End of Zip archive, footer length: 22


```
Exif Image Height : 1104
Exif Pixels Per Unit X : 5669
Exif Pixels Per Unit Y : 5669
Exif Pixel Units : meters
Exif XMP Toolkit : XMP Core 5.4.0
Exif Apple Data Offsets : (Binary data 28 bytes, use -b option to extract)
Warning : [minor] Trailer data after PNG IEND chunk
Image Size : 594x1104
Megapixels : 0.656
```

Binwalk

```
(kali㉿kali)-[~/Desktop]
└─$ unzip dolls.jpg
Archive:  dolls.jpg
warning [dolls.jpg]:  272492 extra bytes at beginning or within zipfile
             (attempting to process anyway)
            inflating: base_images/2_c.jpg
```

```
(kali㉿kali)-[~/Desktop/base_images]
└─$ binwalk 2_c.jpg
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	PNG image, 526 x 1106, 8-bit/color ^I RGBA, non-interlaced
3226	0xC9A	TIFF image data, big-endian, offset of first image director y: 8
187707	0x2DD3B	Zip archive data, at least v2.0 to extract, compressed size : 196043, uncompressed size: 201445, name: base_images/3_c.jpg
383805	0x5DB3D	End of Zip archive, footer length: 22
383916	0x5DBAC	End of Zip archive, footer length: 22

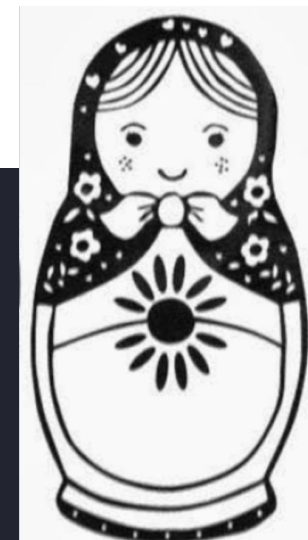


Binwalk

```
(kali@kali)-[~/Desktop/base_images/base_images]
$ ls
3_c.jpg

(kali@kali)-[~/Desktop/base_images/base_images]
$ binwalk 3_c.jpg
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	PNG image, 428 x 1104, 8-bit/color RGBA, non-interlaced
3226	0xC9A	TIFF image data, big-endian, offset of first image directory:
123606	0x1E2D6	Zip archive data, at least v2.0 to extract, compressed size:
essed size: 79807, name: base_images/4_c.jpg		
201423	0x312CF	End of Zip archive, footer length: 22



Binwalk

```

(kali㉿kali)-[~/Desktop/base_images/base_images/base_images]
$ unzip 4_c.jpg
Archive: 4_c.jpg
warning [4_c.jpg]: 79578 extra bytes at beginning or within zipfile
(attempting to process anyway)
  inflating: flag.txt

(kali㉿kali)-[~/Desktop/base_images/base_images/base_images]
$ ls
4_c.jpg  flag.txt

(kali㉿kali)-[~/Desktop/base_images/base_images/base_images]
$ cat flag.txt
picoCTF{bf6acf878dcbd752f4721e41b1b1b66b}
  
```



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